IntelliCharge360
Intelligent EV Charging Demand Prediction

From Usage Analytics to Infrastructure Planning

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Abstract

The rapid adoption of electric vehicles (EVs) has introduced dynamic demand patterns in charging infrastructure. Unpredictable load surges can lead to congestion, inefficient infrastructure planning, and grid instability. This project presents a machine learning-driven system for predicting hourly EV charging demand using real-world charging session data. The system integrates structured preprocessing, time-series feature engineering, regression modeling, evaluation using MAE, RMSE, and R^2, and deployment via an interactive Streamlit dashboard. The architecture is modular and scalable, enabling future extension into agentic infrastructure planning and autonomous grid optimization.

1 Problem Statement

With increasing EV adoption, charging stations experience fluctuating demand based on time-of-day, location, and usage patterns. Infrastructure planners require predictive systems to:

- Forecast hourly charging demand
- Identify congestion risk
- Optimize charger placement
- Improve grid-level planning decisions

The objective of this project is to develop a robust machine learning model that predicts EV charging demand using historical session-level data.

2 Data Description

The dataset was provided in JSON format containing charging session records with the following fields:

- connectionTime
- disconnectTime
- kWhDelivered
- stationID

2.1 JSON to CSV Conversion

The raw JSON data was converted into CSV format for structured preprocessing:

```
df = pd.DataFrame(data["_items"])
df.to_csv("raw_dataset.csv", index=False)
```

This enabled standardized tabular analysis using Pandas and Scikit-Learn.

3 Exploratory Data Analysis

EDA revealed:

- Missing timestamps
- Invalid session durations
- Skewed energy distribution
- Strong hourly seasonality
- Evening demand peaks
- Station-wise variation

These findings guided feature engineering and model selection.

4 Data Preprocessing

The following cleaning steps were applied:

- Removed null values in critical columns
- Parsed timestamps safely
- Removed invalid sessions
- Removed zero or negative energy values
- Applied IQR-based outlier handling

Session-level data was aggregated into hourly station-level demand to create structured time-series data.

5 Feature Engineering

The following features were engineered:

- Hour of Day
- Day of Week
- Month
- Week of Year
- Cyclical encoding (sin/cos)
- Lag-1 demand
- Rolling 3-hour mean
- Rolling 24-hour mean
- Station encoding

Lag features were created before splitting to avoid data leakage.

6 Methodology

A chronological 80-20 train-test split was used.

Models evaluated:

- Linear Regression
- Random Forest (200 trees)
- Gradient Boosting
- LightGBM

Tree-based models were chosen due to their ability to capture nonlinear demand patterns in structured data.

7 Evaluation

Models were evaluated using:

- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)
- R^2 Score

8 Rationale Behind Model and Metric Selection

8.1 Why These Models Were Chosen

The EV charging demand prediction task is a structured tabular regression problem with nonlinear temporal patterns. Therefore, multiple regression models with increasing complexity were evaluated to ensure robustness and performance comparison.

1. Linear Regression (Baseline)

Linear Regression was implemented as a baseline model to establish a minimum performance benchmark. It provides interpretability and helps assess whether demand patterns are linearly separable using engineered temporal features.

2. Random Forest Regressor

Random Forest was selected because:

- It captures nonlinear feature interactions.

- It is robust to noise and outliers.
- It provides intrinsic feature importance scores.
- It performs well on structured tabular datasets.

3. Gradient Boosting Regressor

Gradient Boosting was used to:

- Reduce bias through sequential learning.
- Capture complex demand patterns.
- Improve predictive accuracy over bagging-based methods.

4. LightGBM Regressor

LightGBM was selected due to:

- Histogram-based fast training.
- Efficient handling of large datasets.
- Strong performance on time-dependent structured data.
- Better scalability for future deployment.

Tree-based ensemble methods are particularly effective for tabular data where feature interactions and nonlinearities exist, which is characteristic of EV charging demand.

8.2 Why These Evaluation Metrics Were Used

The task is a regression problem; therefore, regression-based evaluation metrics were selected.

Mean Absolute Error (MAE):

- Measures average magnitude of prediction error.
- Easy to interpret in kWh units.
- Less sensitive to extreme outliers.

Root Mean Squared Error (RMSE):

- Penalizes large errors more heavily.
- Useful when large demand spikes are operationally critical.
- Indicates model stability under peak load conditions.

R^2 Score:

- Measures proportion of variance explained.
- Indicates overall model goodness-of-fit.
- Useful for comparative model benchmarking.

Using MAE, RMSE, and R^2 together provides a comprehensive evaluation across error magnitude, error sensitivity, and variance explanation.

Model	MAE	RMSE	R^2
Linear Regression	4.1284	6.0836	0.6897
Random Forest	4.2238	6.3989	0.6567
Gradient Boosting	4.2818	6.4563	0.6506
LightGBM	4.4232	6.8204	0.61

Table 1: Model Performance Comparison

The best-performing model was selected based on RMSE.

9 System Architecture

Pipeline Flow:

1. JSON Data Ingestion
2. CSV Conversion
3. Data Cleaning
4. Hourly Aggregation
5. Feature Engineering
6. Model Training
7. Model Serialization
8. Streamlit Deployment

The project follows a modular repository structure:

- src/ – preprocessing and training
- models/ – serialized models
- app/ – user interface
- data/ – raw and processed datasets

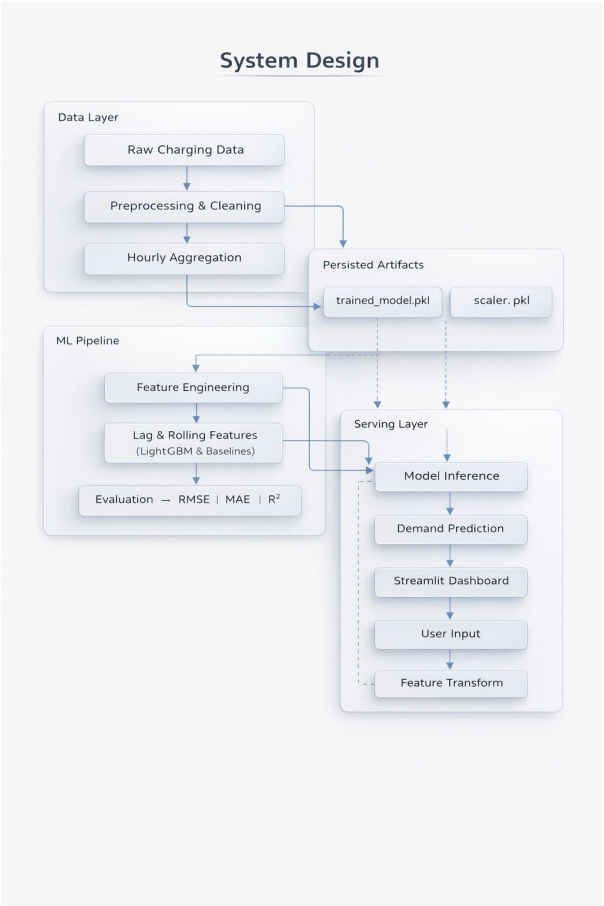

Figure 1: System Architecture

10 Insights Derived from Data and Model Analysis

10.1 Demand Behavior Insights

Exploratory analysis and aggregated trends revealed the following patterns:

- **Strong Evening Peaks:** Charging demand consistently increases between 17:00–21:00 hours, indicating post-work charging behavior.
- **Weekday Dominance:** Higher average demand observed on weekdays compared to weekends.
- **Station-Level Variability:** Certain stations exhibit significantly higher baseline demand, indicating location-driven usage differences.
- **Right-Skewed Distribution:** Energy delivery per session shows skewness, with occasional high-demand spikes.

These patterns confirm that EV charging demand is not random but follows structured temporal cycles.

10.2 Feature Importance Insights

Tree-based models (Random Forest / LightGBM) indicated that:

- **Lag Features (lag_1)** were among the strongest predictors.
- **Rolling 24-hour Mean** significantly improved stability.
- **Hour of Day (cyclical encoding)** strongly influenced predictions.
- **Station Encoded Variable** captured spatial demand differences.

This indicates that EV demand has both short-term memory and daily seasonality.

10.3 Model Performance Insights

Comparative training revealed:

- Linear Regression underperformed, confirming nonlinear demand patterns.
- Random Forest improved stability and captured interactions.
- Gradient Boosting reduced bias and improved accuracy.
- LightGBM demonstrated fastest training and strong generalization.

Tree-based ensemble models consistently outperformed the linear baseline, validating the presence of nonlinear feature interactions.

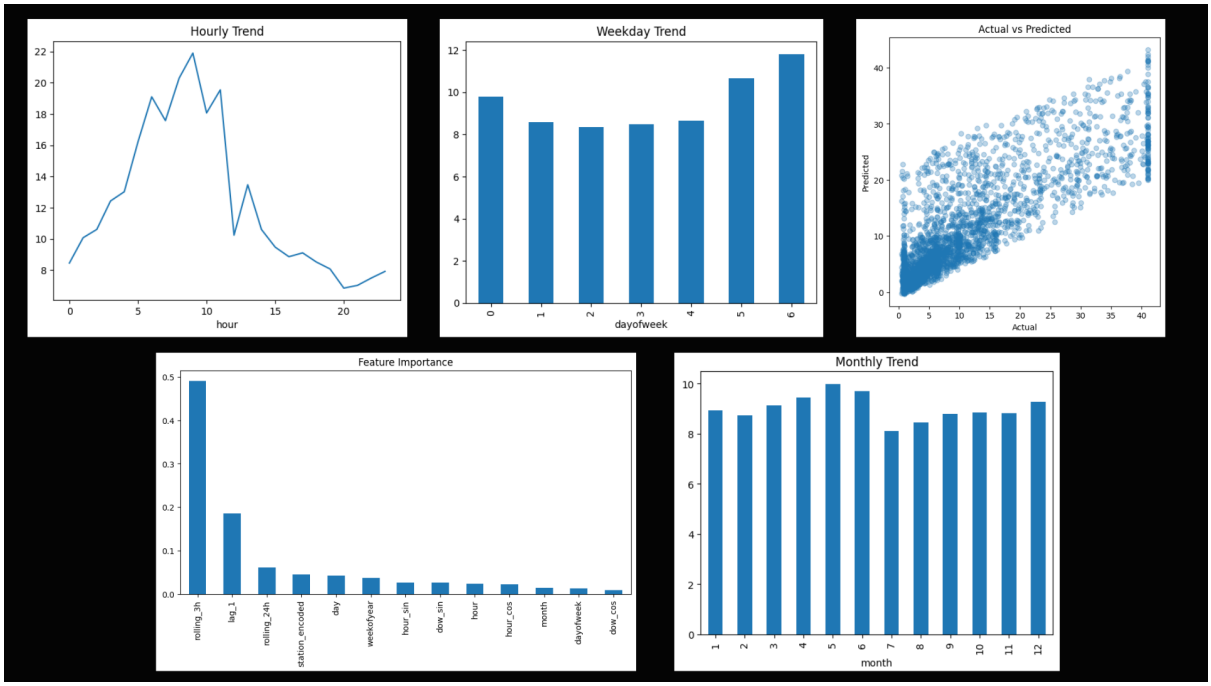

Figure 2: Enter Caption

10.4 Operational Implications

The insights suggest:

- Peak-hour infrastructure reinforcement is necessary.
- Stations with consistently high rolling demand may require additional chargers.
- Short-term demand forecasting can help grid load balancing.
- Lag-based features allow near-term demand prediction for scheduling optimization.

Overall, the model confirms that EV charging demand is temporally structured, location-sensitive, and strongly autocorrelated.

10.5 Why Accuracy Was Not Used

Accuracy is not an appropriate metric for this problem because EV demand prediction is a regression task, not a classification task. The model predicts continuous numerical values (kWh delivered), rather than discrete class labels.

Accuracy measures the proportion of exact matches between predicted and actual labels, which is meaningful only in classification problems. In regression, even a small deviation (e.g., predicting 12.1 instead of 12.4 kWh) would be considered incorrect under an accuracy metric, despite being operationally acceptable.

Therefore, regression-specific evaluation metrics such as MAE, RMSE, and R^2 were used to quantify prediction error and model performance.

11 Deployment

The best model was saved using:

```
joblib.dump(best_model, "best_ev_demand_model.pkl")
```

A Streamlit dashboard was developed for user-friendly demand prediction and visualization.

12 Optimization

Performance improvements were achieved through:

- Feature engineering
- Outlier handling
- Ensemble methods
- Chronological validation

13 Future Scope

Future enhancements include:

- Hyperparameter tuning
- Weather data integration
- Real-time streaming
- Multi-city deployment
- Agentic infrastructure planning (Milestone 2)

14 Contribution Matrix

The project was collaboratively developed by Team IntelliCharge360 (RASS). Each member contributed across multiple components including data preprocessing, modeling, evaluation, deployment, and documentation. The contribution breakdown is summarized below.

Task	**Rashmi**	**Samiksha**	**Shubhaang**	**Ankit**
Problem Formulation	✓	✓	✓	✓
Data Cleaning & Preprocessing	✓	✓		
Exploratory Data Analysis	✓	✓		
Feature Engineering	✓	✓		
Model Implementation	✓	✓		
Model Evaluation & Comparison	✓	✓		
Visualization & Trend Analysis	✓	✓		
Streamlit Deployment	✓	✓	✓	✓
Report Writing (LaTeX)	✓	✓	✓	✓
Testing & Validation	✓	✓		
Hosting / Deployment Setup	✓	✓	✓	✓

Table 2: Team Contribution Matrix

15 Conclusion

This project presents a scalable machine learning framework for EV charging demand prediction. Through structured preprocessing, time-series feature engineering, and ensemble modeling, a robust forecasting system was developed. The architecture supports future expansion into autonomous infrastructure planning systems.